

LABOUR MARKET
ANALYTICS :
A PLFS DATA
ANALYSIS
USING PYTHON

Labour Market Analytics

A PLFS Data Analysis Using Python

1. Project Overview

This project focuses on analyzing the Periodic Labour Force Survey (PLFS) 'Persons Surveyed' dataset using Python to extract meaningful insights about India's labour force survey coverage. The analysis includes data cleaning, feature engineering, exploratory data analysis (EDA), statistical analysis, and visualization to understand patterns and trends in survey coverage across states, years, and quarters.

The project aims to transform raw government survey data into structured, decision-ready information that supports a better understanding of India's labour market data collection framework.

2. Data Source

- Dataset Name: PLFS Persons Surveyed
- Source: Periodic Labour Force Survey (PLFS), Ministry of Statistics and Programme Implementation (MoSPI), Government of India — via India Data Portal (Open Government Data)
- File Type: CSV
- Number of Records: 529
- Number of Attributes: 24 (original)
- Location: India (22 states/UTs + All-India aggregate)
- Year/Timeline: 2018 to 2024 (quarterly)

The dataset contains PLFS survey-coverage information such as the number of blocks surveyed, households surveyed, and the age-and-gender demographic breakdown of persons surveyed in each state, quarter, and year.

3. Problem Statement / Business Problem

Understanding labour market trends is essential for assessing employment conditions and supporting policy decisions. Organizations and policymakers collect large volumes of PLFS survey data but often struggle to extract structured insights from it. This project aims to analyze PLFS survey-coverage data to identify workforce sampling patterns, demographic representation, and state/quarter-wise differences using Python-based data analytics.

4. Objectives

1. To analyze year-wise trends (2018–2024) in the number of persons, households, and blocks surveyed to identify changes in survey coverage over time.

2. To compare state-wise survey coverage by identifying states with the highest and lowest numbers of surveyed households and persons.
3. To examine quarter-wise (seasonal) survey coverage across the four PLFS survey periods (April–June, July–September, October–December, and January–March).
4. To analyze the demographic distribution of surveyed persons by age groups and gender (Male/Female).
5. To identify state-wise and year-wise patterns, trends, and outliers in survey coverage using comparative analysis and visualizations.
6. To derive meaningful insights and present the findings through clear visualizations, supporting a better understanding of PLFS survey coverage and data distribution.

5. Attribute (Column / Feature) Details

Attribute Name	Data Type	Description
Year	Text	Survey year.
Quarter	Text	Quarter during which the survey was conducted.
state_name	Text	Name of the state or union territory.
num_blocks_surveyed	Numeric	Number of survey blocks covered.
Households	Numeric	Total number of households surveyed.
males_age_0_4_years	Numeric	Number of males aged 0–4 years.
females_age_0_4_years	Numeric	Number of females aged 0–4 years.
persons_age_0_4_years	Numeric	Total number of persons aged 0–4 years.
males_age_5_14_years	Numeric	Number of males aged 5–14 years.
females_age_5_14_years	Numeric	Number of females aged 5–14 years.
persons_age_5_14_years	Numeric	Total number of persons aged 5–14 years.
males_age_15_29_years	Numeric	Number of males aged 15–29 years.
females_age_15_29_years	Numeric	Number of females aged 15–29 years.
persons_age_15_29_years	Numeric	Total number of persons aged 15–29 years.
males_age_15_59_years	Numeric	Number of males aged 15–59 years.
females_age_15_59_years	Numeric	Number of females aged 15–59 years.
persons_age_15_59_years	Numeric	Total number of persons aged 15–59 years.
males_age_15_above_years	Numeric	Number of males aged 15 years and above.
females_age_15_above_years	Numeric	Number of females aged 15 years and above.
persons_age_15_above_years	Numeric	Total number of persons aged 15 years and above.
males_all_ages	Numeric	Total number of males across all age groups.
females_all_ages	Numeric	Total number of females across all age groups.
persons_all_ages	Numeric	Total number of persons across all age groups.
state_code	Text	Unique code assigned to the state or union territory.

6. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

- Google Colab
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7. Data Pre-Processing and Feature Engineering

The dataset was prepared for analysis using standard data preprocessing techniques.

Data Cleaning Steps

- Checked data types of all columns
- Identified and verified missing values (none found)
- Checked and removed duplicate records (none found)
- Standardized text fields such as state names and quarter labels
- Converted count columns (households, persons, etc.) from float to integer type

Feature Engineering

New features were created to improve analysis and enable deeper demographic and time-based insights.

- `is_all_india` — flag to separate the All-India aggregate row from individual states
 - `period_date` / `fiscal_quarter` / `period_label` — proper time features built from year and quarter
 - `persons_per_household`, `households_per_block`, `persons_per_block` — survey intensity metrics
 - `sex_ratio_all_ages` — females per 1000 males
 - `share_working_age_15_59` — % of surveyed persons in the working-age (15–59) group
 - `male_share_pct` / `female_share_pct` — gender composition of surveyed persons
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8. Analysis and Visualizations

The dataset was analyzed using Exploratory Data Analysis techniques to understand patterns and relationships among survey coverage and demographic variables.

9. Exploratory Data Analysis (EDA)

Initial inspection included:

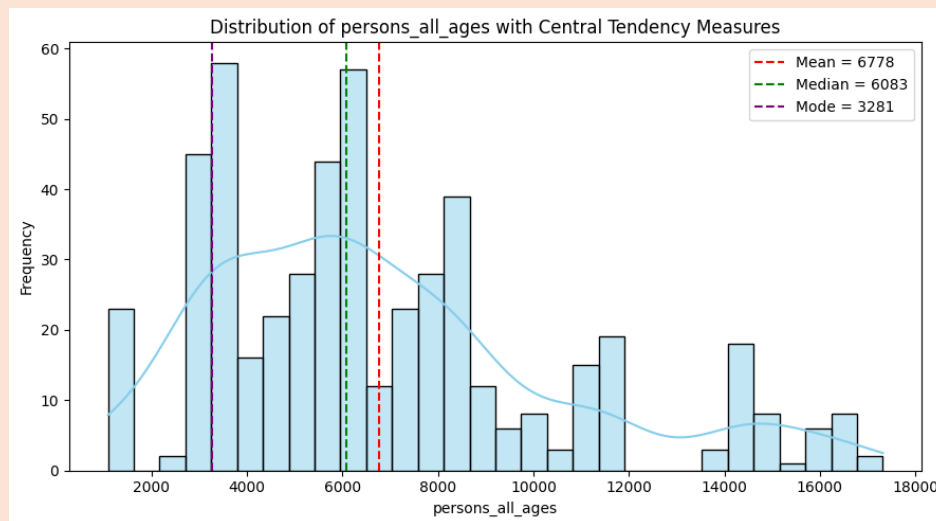
- Data frame structure verification (shape, columns, dtypes)
- Statistical summary of numerical columns
- Null value verification
- Duplicate record validation

EDA helped in understanding the distribution and characteristics of PLFS survey coverage data across states, years, and quarters.

10. Statistical Analysis

Statistical measures were used to summarize the persons_all_ages variable (total persons surveyed) across states and periods.

Measures of Central Tendency



Interpretation:

1. The mean (5939) is slightly higher than the median (5786), indicating that the distribution is positively (right) skewed. This suggests that a few states or observations with very high numbers of surveyed persons are increasing the average.
2. The mode (3281) is much lower than both the mean and median, showing that the most frequently occurring value is lower than the overall average. This indicates that many states have relatively smaller survey counts, while fewer states have much larger counts.
3. The histogram shows that the data is not perfectly symmetric and exhibits variability across observations. The spread of values and the long right tail indicate differences in survey coverage among states, with some states having substantially higher numbers of persons surveyed than others.

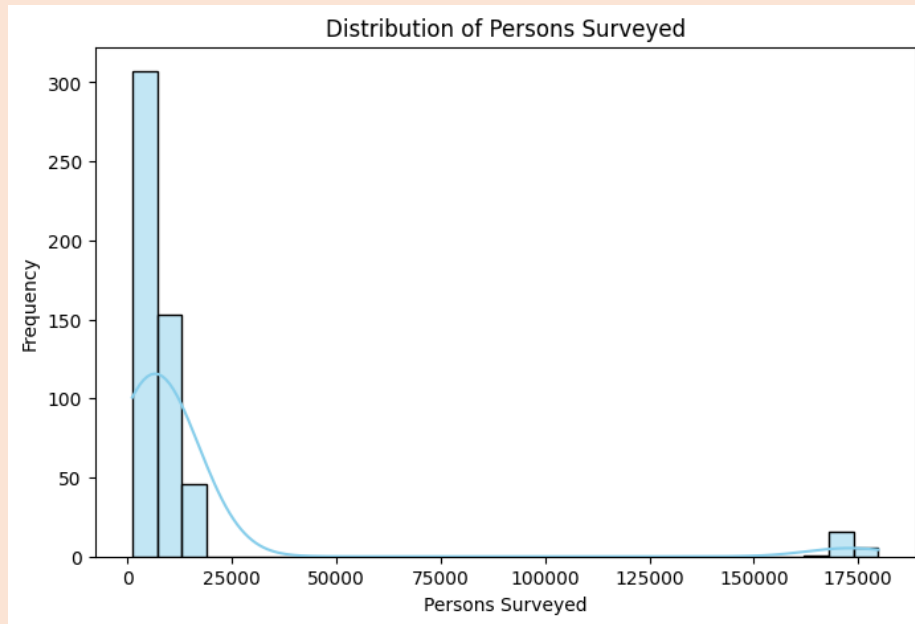
11. Data Visualization

Visualization techniques were used to represent PLFS survey data graphically and make coverage and demographic patterns easier to understand.

1. Univariate Analysis

Univariate analysis was performed to analyze one variable at a time, understanding its distribution, central tendency, and variability.

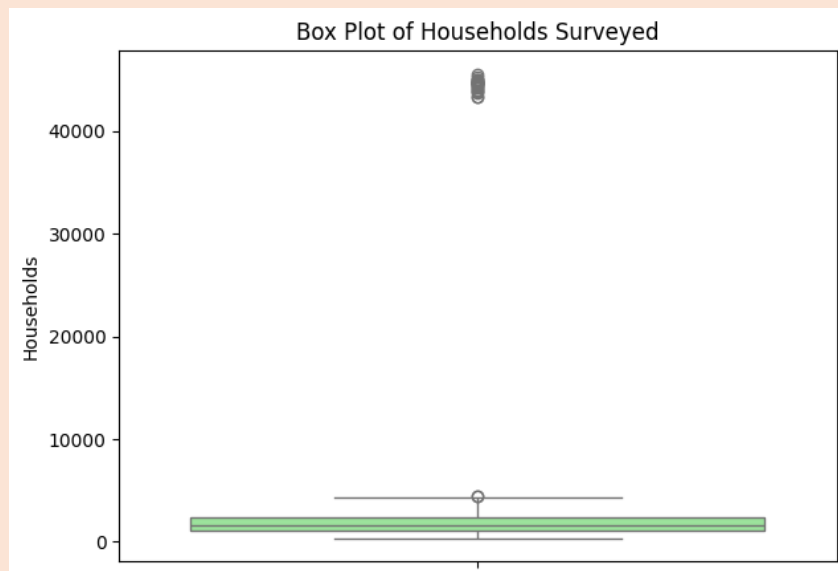
Histogram – Distribution of Persons Surveyed



Interpretation:

1. Most survey records contain fewer than 20,000 persons surveyed.
2. The data is not normally distributed because of the long right tail.
3. The extreme values suggest the presence of national summary (All-India) records, which are much larger than state-level records.
4. These outliers increase the average (mean), making it higher than the value typically observed for most states

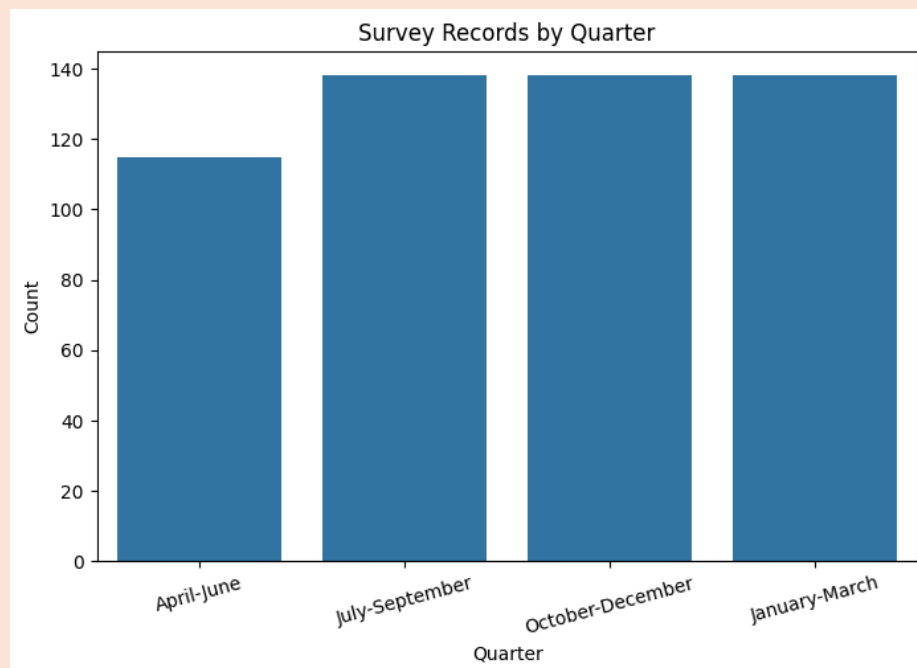
Box Plot – Households Surveyed



Interpretation:

1. The median (middle line inside the box) is around 1,500–1,800 households, indicating that about half of the survey observations include fewer households than this value and half include more.
2. The box (Interquartile Range - IQR) is relatively narrow, showing that 50% of the survey observations are concentrated within a moderate range. This indicates fairly consistent survey coverage across most states and quarters.
3. The lower whisker extends close to 0 households, suggesting that some survey units had relatively small sample sizes.
4. The upper whisker reaches approximately 4,000–4,500 households, representing the highest values that are not considered unusual.
5. A large number of points above the upper whisker (around 43,000–45,000 households) are identified as outliers. These observations are far higher than the rest of the data.

Count Plot – Survey Quarter



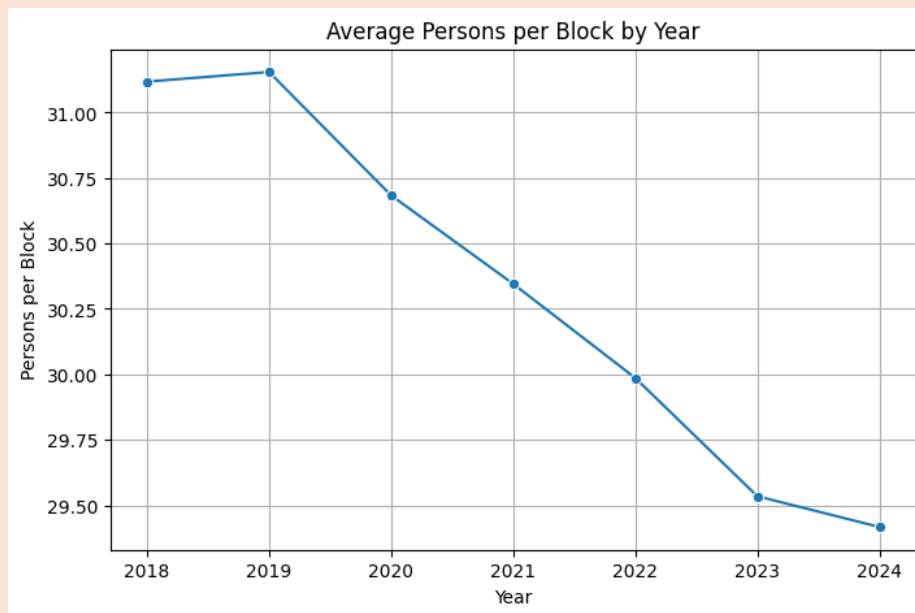
Interpretation:

1. July–September, October–December, and January–March have the highest and nearly identical number of survey records (about 138 each).
2. April–June has the lowest number of records (about 115).
3. The dataset is well distributed across quarters, with only a slight reduction in April–June.
4. The balanced distribution supports reliable quarter-wise comparisons and seasonal trend analysis.

2. Bivariate Analysis

Bivariate analysis was performed to study the relationship between two variables to identify trends, associations, and correlations.

Line Plot – Year vs Persons per Block



Interpretation:

1. The average persons per block remained highest in 2018 and 2019, at approximately 31.1 persons per block.
2. From 2020 onwards, the average declined gradually each year, indicating a consistent downward trend in the number of persons surveyed per block.
3. By 2022, the average had decreased to around 30 persons per block.
4. The lowest averages were observed in 2023 (29.5) and 2024 (29.4).
5. Overall, the average decreased by approximately 1.7 persons per block between 2018 and 2024, suggesting a modest but steady reduction in survey density.

Scatter Plot – Male vs Female Surveyed Persons by Age Group

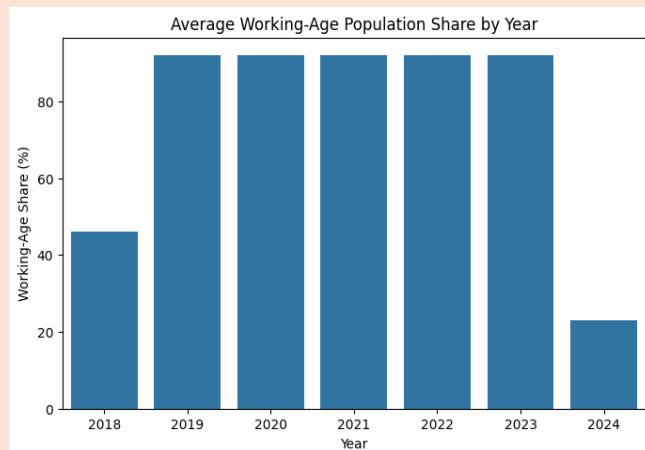


Interpretation:

This scatter plot compares the number of male surveyed persons (X-axis) with the number of female surveyed persons (Y-axis) across different age groups.

1. The points form an upward trend, indicating a strong positive relationship between male and female surveyed persons.
2. As the number of male respondents increases, the number of female respondents also increases across the age groups.
3. The point representing the 15+ years age group is the farthest from the origin, indicating the largest number of surveyed persons for both males and females.
4. The 15–59 years age group also has a high concentration of surveyed persons, reflecting the importance of the working-age population in the PLFS survey.
5. The 0–4 years age group is closest to the origin, indicating the smallest number of surveyed persons. The points are distributed close to an imaginary diagonal line, suggesting that male and female survey participation is proportionally balanced across all age groups.

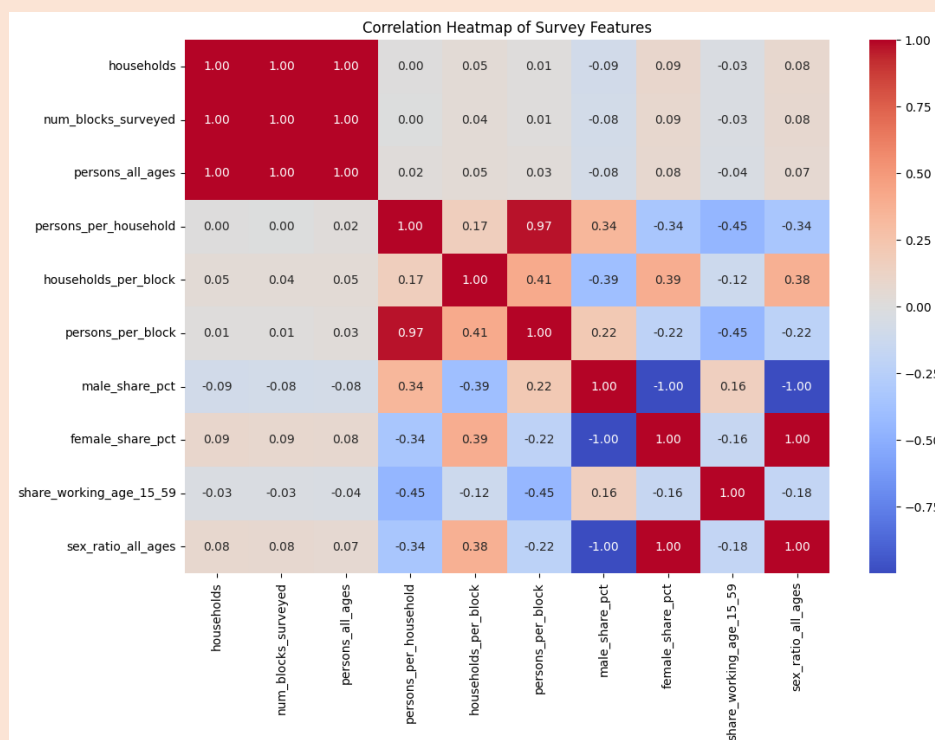
Bar Plot – Working-Age Population Share by Year



Interpretation – Average Working-Age Population Share by Year :

1. The bar chart shows the average share of the working-age population (15–59 years) in the surveyed population from 2018 to 2024.
2. The working-age population share remained consistently high (around 92%) from 2019 to 2023, indicating that the majority of surveyed persons belonged to the economically active age group.
3. 2018 recorded a much lower share of approximately 46%, while 2024 dropped further to about 23%.
4. The sharp decline in 2018 and 2024 contrasts with the stable pattern observed between 2019 and 2023.
5. Such large changes are unusual for population demographics, as the proportion of the working-age population typically changes gradually over time rather than abruptly.

Correlation Heatmap



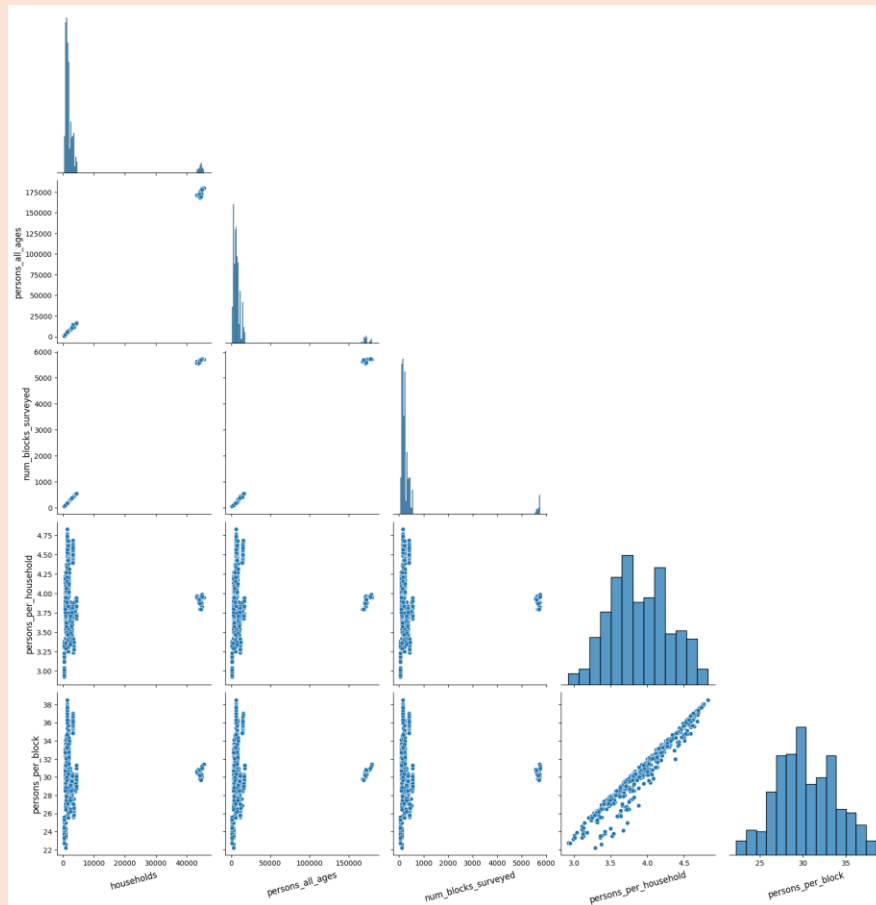
Interpretation:

1. Households, blocks surveyed, and persons surveyed are almost perfectly correlated (1.00), confirming consistent survey expansion.
2. Persons per household and persons per block are strongly related (0.97), indicating household size influences survey density.
3. Male share and female share have a perfect negative correlation (-1.00) because they are complementary proportions.
4. Year has little or no correlation with other variables, suggesting that survey coverage remained stable over time.
5. Most correlations align with the way the PLFS data is collected and the way some derived features are calculated.

3. Multivariate Analysis

Multivariate analysis was used to examine the relationship among three or more variables simultaneously using pair plots, grouped analysis, pivot tables, heatmaps, and advanced visualizations.

Pairplot (Households, Persons, Blocks, Persons per Household, Persons per Block)



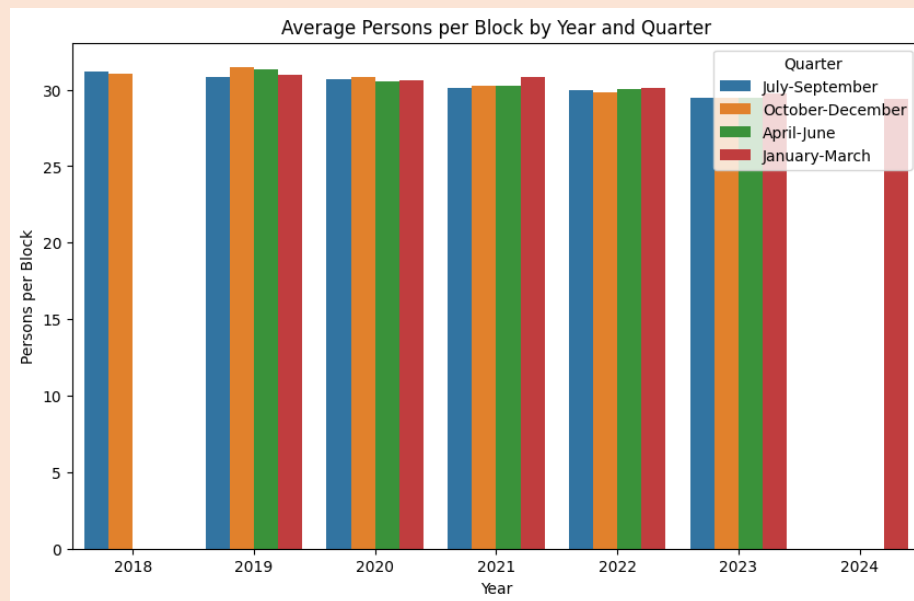
The pair plot provides a comprehensive view of the distribution of each survey variable (diagonal histograms) and the relationships between pairs of variables (scatter plots). It helps identify patterns, trends, correlations, and potential outliers in the PLFS dataset.

Interpretation :

1. The diagonal histograms show that households, persons surveyed, and blocks surveyed are right-skewed, indicating that most survey observations have moderate values, while a few observations (such as All-India aggregates) have much larger values.
2. The scatter plots between households, persons surveyed, and blocks surveyed display a strong positive linear relationship. As the number of surveyed blocks increases, the number of households and persons surveyed also increases.
3. Persons per household is concentrated between 3.0 and 4.8, showing relatively little variation across survey observations.

4. There is a strong positive relationship between persons per household and persons per block, indicating that blocks with larger average household sizes also tend to have more surveyed persons.
5. A few observations are clearly separated from the main cluster, representing large aggregate records (such as All-India totals) or other high-value observations.

Grouped Analysis (Year × Quarter × Persons per Block)

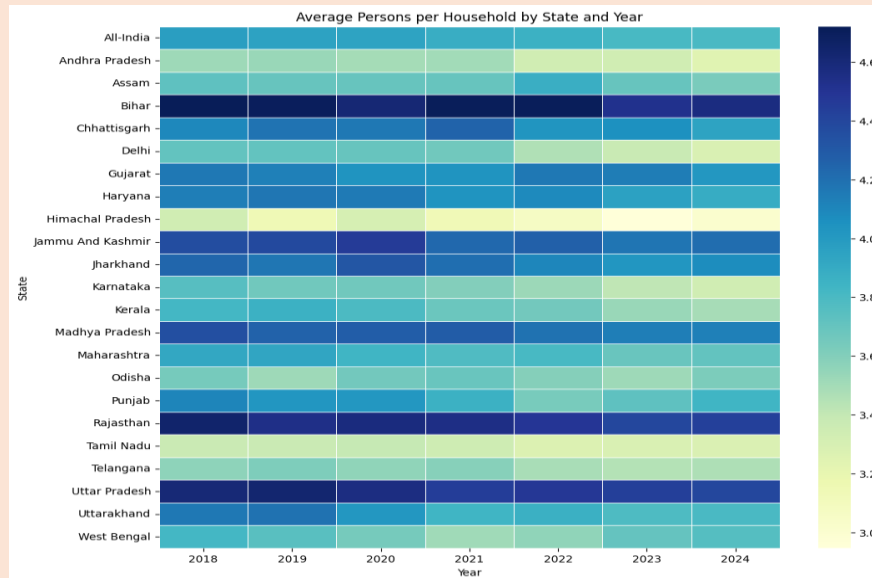


Interpretation – Average Persons per Block by Year and Quarter

The grouped bar chart compares the average number of persons surveyed per block across different years (2018–2024) and PLFS survey quarters.

1. The average number of persons surveyed per block remained around 30–31 persons throughout the study period, indicating a stable survey design.
2. 2018 and 2019 recorded the highest averages, with values slightly above 31 persons per block across most quarters.
3. From 2020 onwards, there is a gradual decline in the average persons surveyed per block across all quarters.
4. By 2023 and 2024, the average decreased to approximately 29–30 persons per block.
5. Within each year, the differences between quarters are very small, suggesting that survey coverage remained consistent across seasonal survey rounds.
6. The January–March quarter is slightly higher than the other quarters in some years (such as 2021 and 2022), but the variation is minimal.

Pivot Table Heatmap (State × Year × Persons per Household)

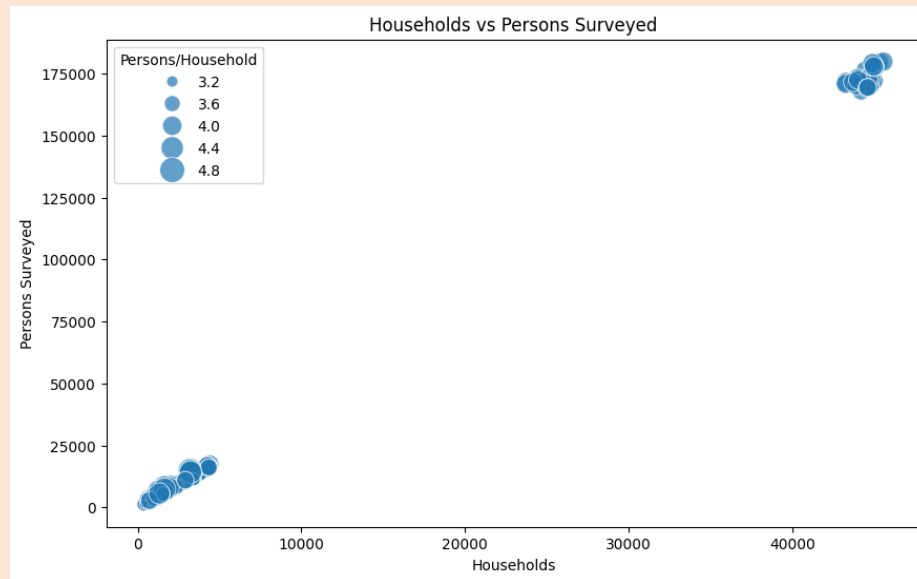


The heatmap illustrates the average number of persons per household across different states from 2018 to 2024. Darker shades indicate higher household sizes, while lighter shades represent smaller household sizes.

Interpretation :

1. The average household size ranges approximately from 3.0 to 4.7 persons per household across the states.
2. Bihar, Uttar Pradesh, and Rajasthan consistently record the highest average household sizes, with values generally above 4.4 persons per household throughout the study period.
3. Himachal Pradesh, Tamil Nadu, and Delhi have comparatively smaller household sizes, averaging around 3.0–3.4 persons per household.
4. Most states show only slight year-to-year variations, indicating that household size remained relatively stable during 2018–2024.
5. A few states, such as Kerala, Punjab, and West Bengal, display a gradual decline in average household size over the years.
6. The All-India average remains close to 4 persons per household, showing only minor changes across the study period.

Bubble Scatter Plot (Households × Persons Surveyed × Persons per Household)

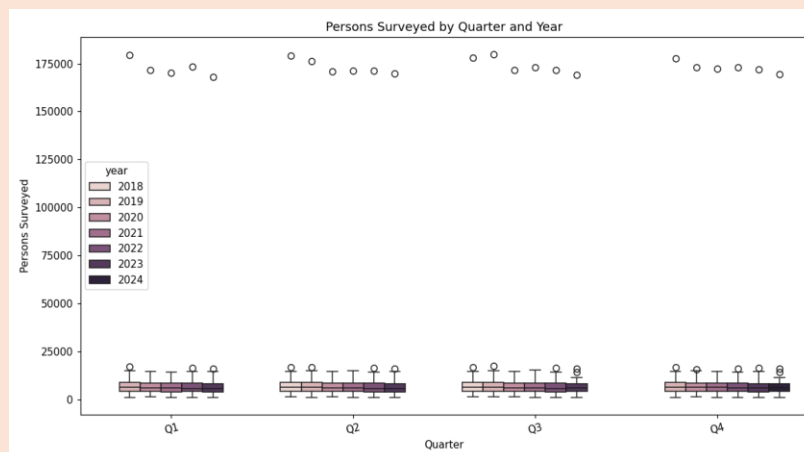


The bubble chart illustrates the relationship between the number of households surveyed (X-axis) and the number of persons surveyed (Y-axis). The bubble size represents the average number of persons per household, providing an additional demographic dimension.

Interpretation:

1. There is a strong positive relationship between the number of households surveyed and the number of persons surveyed. As the number of households increases, the number of persons surveyed also increases proportionally.
2. The chart shows two distinct clusters: Lower-left cluster: Represents individual state-level survey records, where household and person counts are relatively smaller. Upper-right cluster: Represents All-India aggregate records, which combine data from all states and therefore have much higher values.
3. The bubble sizes are fairly similar across observations, indicating that the average persons per household remains relatively stable, generally between 3 and 5 persons.
4. No major deviations from the upward trend are observed, suggesting that the survey data follows a consistent sampling pattern.

Grouped Box Plot (Quarter × Persons Surveyed × Year)

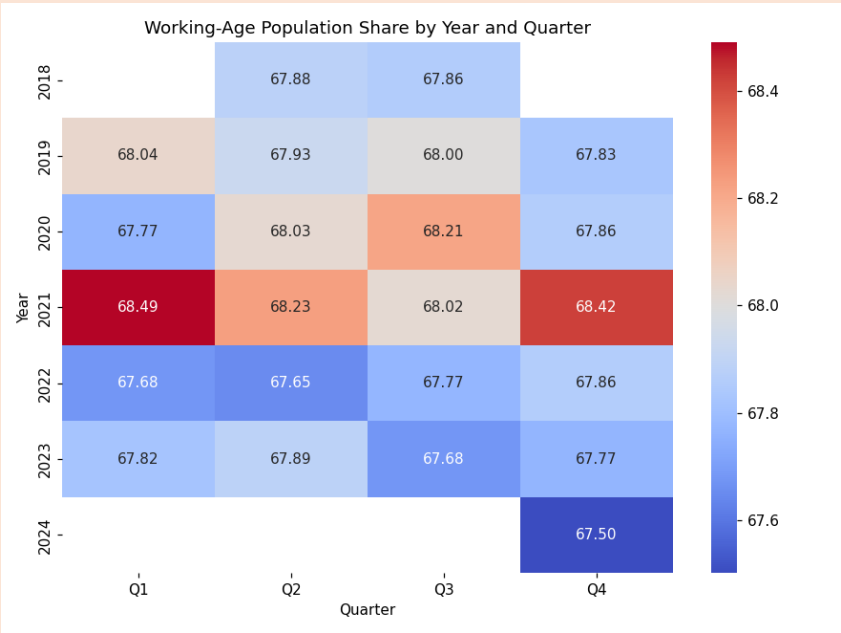


The box plot compares the distribution of persons surveyed across the four PLFS survey quarters (Q1–Q4) for each year from 2018 to 2024.

Interpretation:

- 1. The median number of persons surveyed is similar across all quarters, indicating that the survey maintained a consistent sample size throughout the year.
- 2. The Interquartile Range (IQR) is relatively narrow in each quarter, showing that most survey observations are concentrated within a similar range and that survey coverage is stable.
- 3. The box plots for different years overlap considerably, suggesting very little variation in the distribution of persons surveyed across years.
- 4. Several extreme outliers appear around 170,000–180,000 persons in every quarter. These represent the All-India aggregate records, which are much larger than individual state-level observations.
- 5. Apart from these aggregate values, most state-level survey records fall within a comparable range, indicating a uniform sampling pattern

Pivot Table Heatmap (Year × Quarter × Working-Age Share)



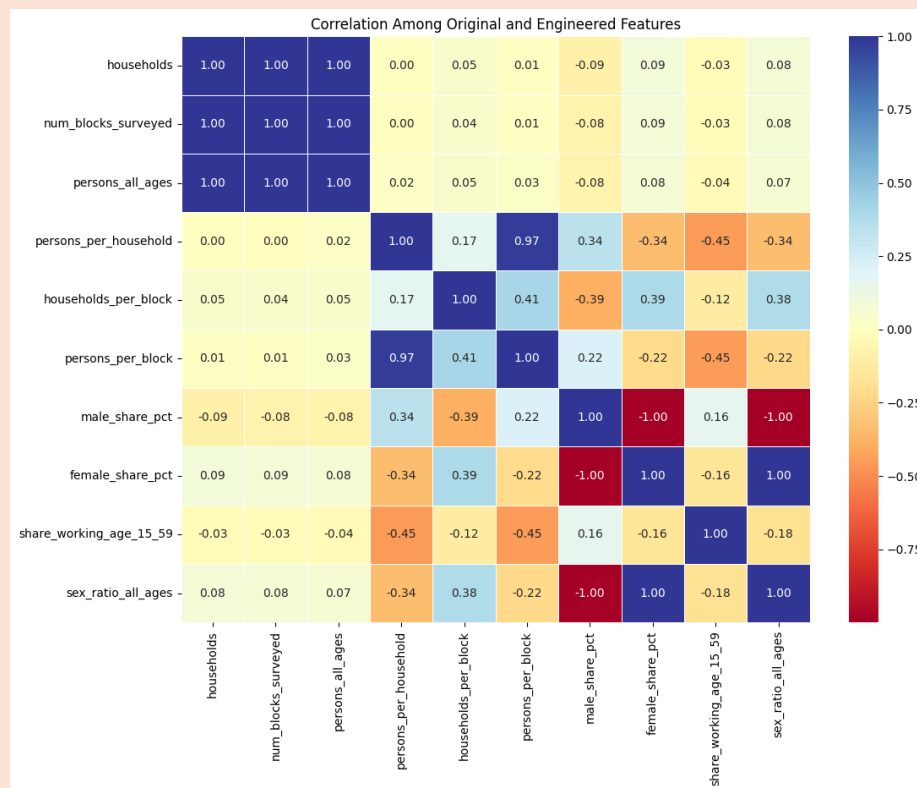
The heatmap presents the average share of the working-age population (15–59 years) across the four PLFS survey quarters (Q1–Q4) from 2018 to 2024. The colour intensity represents the percentage of the working-age population, with warmer colours indicating higher shares and cooler colours indicating lower shares.

Interpretation:

- 1. The working-age population share remains highly consistent, ranging from approximately 67.5% to 68.5% across all years and quarters.
- 2. 2021 records the highest working-age population share, particularly in Q1 (68.49%) and Q4 (68.42%), indicating a slightly higher proportion of working-age individuals during that year.

- 2024 Q4 has the lowest value (67.50%), although the difference from other years is very small.
- Across each year, the variation between quarters is minimal, suggesting that the demographic composition of the surveyed population remains stable throughout the survey cycle.
- The overall difference between the highest and lowest values is less than one percentage point, indicating remarkable consistency in the survey population.

Correlation Heatmap (Original + Engineered Features)



Interpretation – Correlation Among Original and Engineered Features

The correlation heatmap illustrates the strength and direction of the relationships between the original PLFS survey variables and the engineered features. Correlation values range from -1 to +1, where: +1 indicates a perfect positive relationship. 0 indicates no linear relationship. -1 indicates a perfect negative relationship.

The colour intensity helps identify the strength of these relationships, with darker blue representing strong positive correlations and darker red representing strong negative correlations.

- Households, blocks surveyed, and persons surveyed are almost perfectly correlated, confirming the consistency of the PLFS sampling framework.
- Persons per household and persons per block have a very strong positive relationship (0.97).
- Male share and female share exhibit a perfect negative correlation (-1.00) because they are complementary proportions.
- Sex ratio is strongly associated with gender-share variables, as expected.
- Working-age population share has only moderate relationships with survey density measures, indicating that demographic composition is relatively stable regardless of survey size.

6. The engineered features enhance the dataset by providing demographic insights while preserving the integrity of the original survey variables.
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12. Summary of Findings

- The number of persons surveyed is right-skewed — most states cluster around moderate survey counts, while a few states/periods show much higher coverage.
 - The working-age population (15–59 years) share remains fairly stable across years and quarters, indicating consistent demographic representation in the survey sample.
 - Male and female shares of surveyed persons are complementary and consistent, reflecting a balanced gender composition in PLFS sampling.
 - Households, persons surveyed, and blocks surveyed are strongly positively correlated — larger survey operations scale together across these variables.
 - Persons-per-household values vary meaningfully across states and years, visible in the state $\times$ year heatmap, pointing to regional differences in household composition.
 - Quarterly and year-wise survey implementation is broadly consistent, with some natural variability that may reflect operational or seasonal factors.
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13. Types of Analysis

1. Descriptive Analysis

Descriptive analysis summarizes historical PLFS survey data.

- Average persons and households surveyed per state/quarter
- Total survey coverage distribution across years
- State-wise and quarter-wise survey counts

2. Diagnostic Analysis

Diagnostic analysis explains why certain patterns occur in the survey data.

- Reasons behind higher survey coverage in specific states
- Factors affecting variation in working-age population share across periods

3. Predictive Analysis

Predictive analysis helps forecast future survey and demographic patterns.

- Forecasting future survey coverage (persons/households) by state
- Predicting demographic composition trends for upcoming PLFS rounds

4. Prescriptive Analysis

Prescriptive analysis suggests actions based on the insights derived.

- Recommending sample allocation adjustments for under-covered states
- Guiding resource planning for future PLFS survey rounds
- Improving consistency of quarterly survey operations

14. Future Enhancements and Suggestions

- Integrate actual PLFS employment/unemployment indicators (LFPR, WPR, Unemployment Rate) to extend this coverage analysis into a full labour market study.
- Apply machine learning models to detect anomalies or irregularities in survey coverage patterns.
- Build interactive dashboards for state $\times$ year $\times$ quarter drill-down analysis.
- Automate data refresh and reporting as new PLFS quarterly releases become available.

15. Conclusion

The project successfully analyzed the PLFS ‘Persons Surveyed’ dataset using Python and extracted useful insights into India’s labour force survey coverage.

Data preprocessing, feature engineering, and visualization helped in understanding demographic patterns and state/quarter-wise variability in survey coverage.

The results of this project can support MoSPI and policymakers in evaluating and improving the design and consistency of future PLFS survey rounds, and lay the groundwork for deeper labour market analysis using complementary PLFS employment indicators.

Python Notebook Interpretation & Interactive Report Enhancements

This revised report integrates interpretations from the accompanying Python notebook. Each visualization should include: Objective, Chart Description, Key Findings, Business Insight, and Recommendation.

Suggested Interactive Layout

- Cover page with modern theme
- Table of Contents
- Code Snapshot + Output + Interpretation on facing sections
- Colored insight boxes
- KPI callout cards
- Summary at end of each analysis section
- Final dashboard and conclusions